Contextualised Embeddings

Dr. Andy Barnes
apb41@bath.ac.uk
9.22 Wessex House

Where are we?

	Lecture A	Lecture B	Lab
Week 1	Introduction to NLP	Language Preprocessing	Genre Classification
Week 2	Document Representations	Handling Sparsity	Sentiment Analysis
Week 3	Static Word Embeddings	$\boxed{\text{Contextual Word Embeddings}}$	Embedding Vis.
Week 4	Information Extraction	Syntax Parsing	Relationship Extr.
Week 5	Classical Sequence Models	Neural Sequence Models	Text Generation

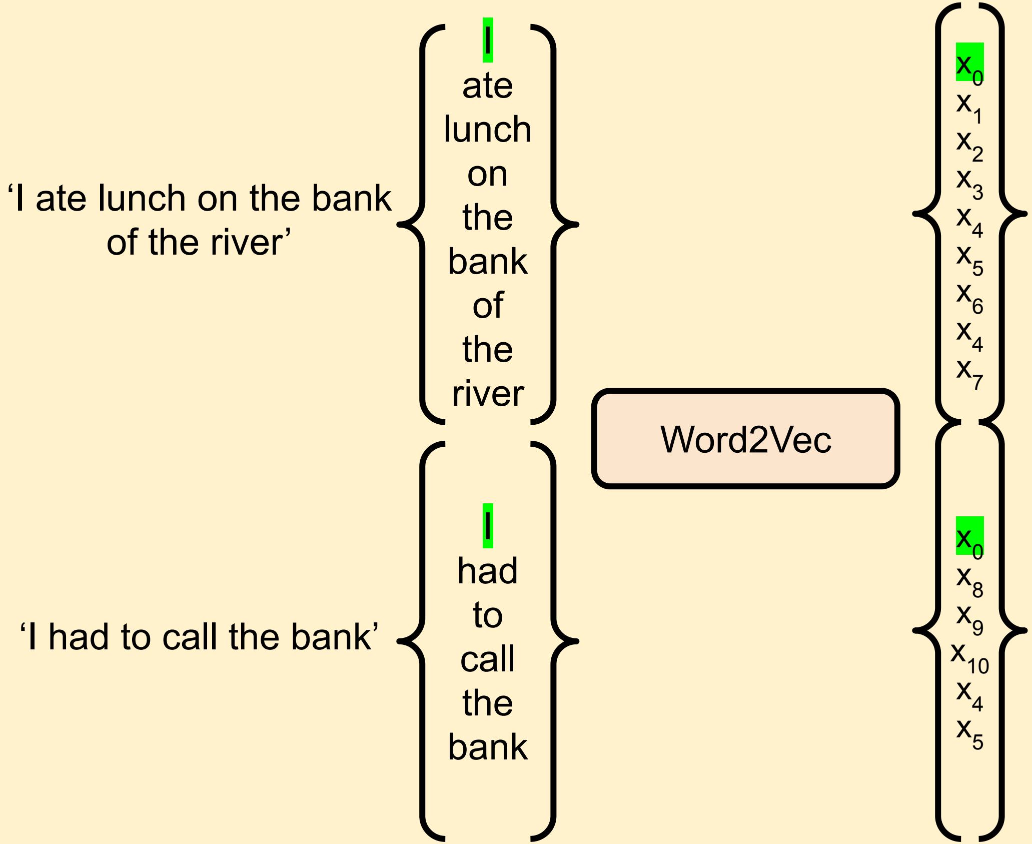
Learning Objectives

Describe the limitations of static embeddings.

Illustrate the architecture of ELMo.

Explain the benefits and drawbacks of pre-trained weights.

Static Embeddings: The Problem



Static Embeddings: The Problem

‘I ate lunch on the bank of the river’

I
ate
lunch
on
the
bank
of
the
river

‘I had to call the bank’

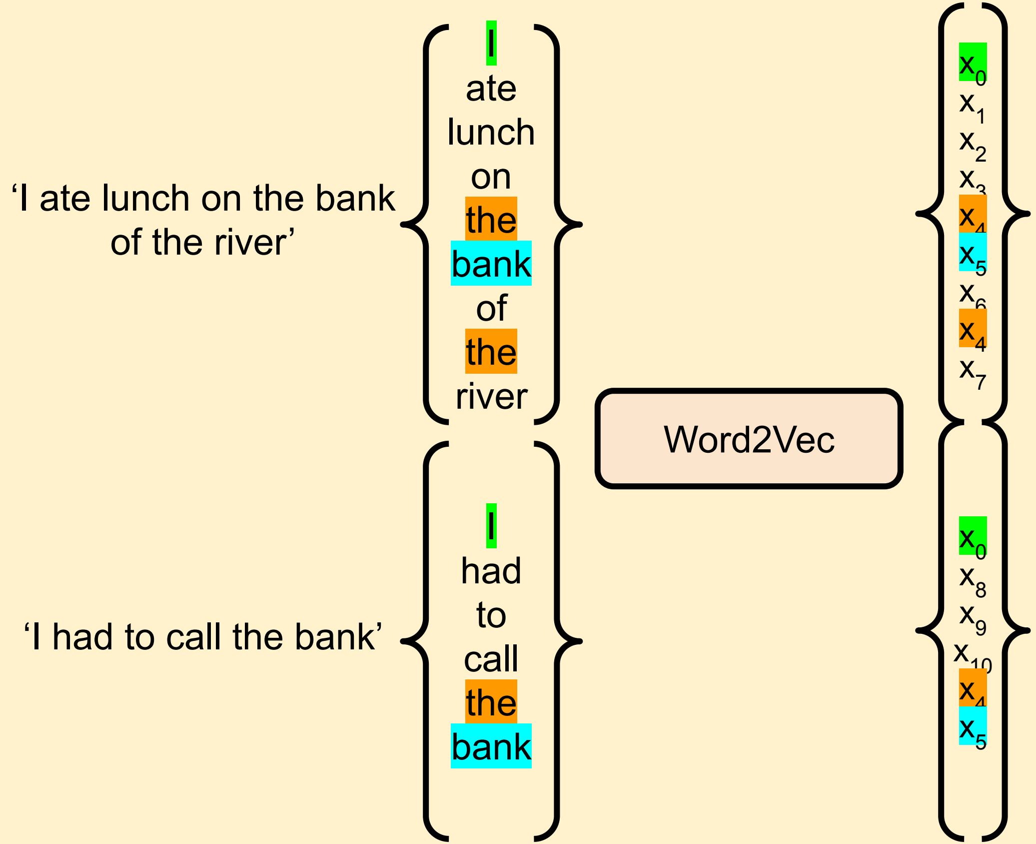
I
had
to
call
the
bank

Word2Vec

x_0
x_1
x_2
x_3
x_4
x_5
x_6
x_4
x_7

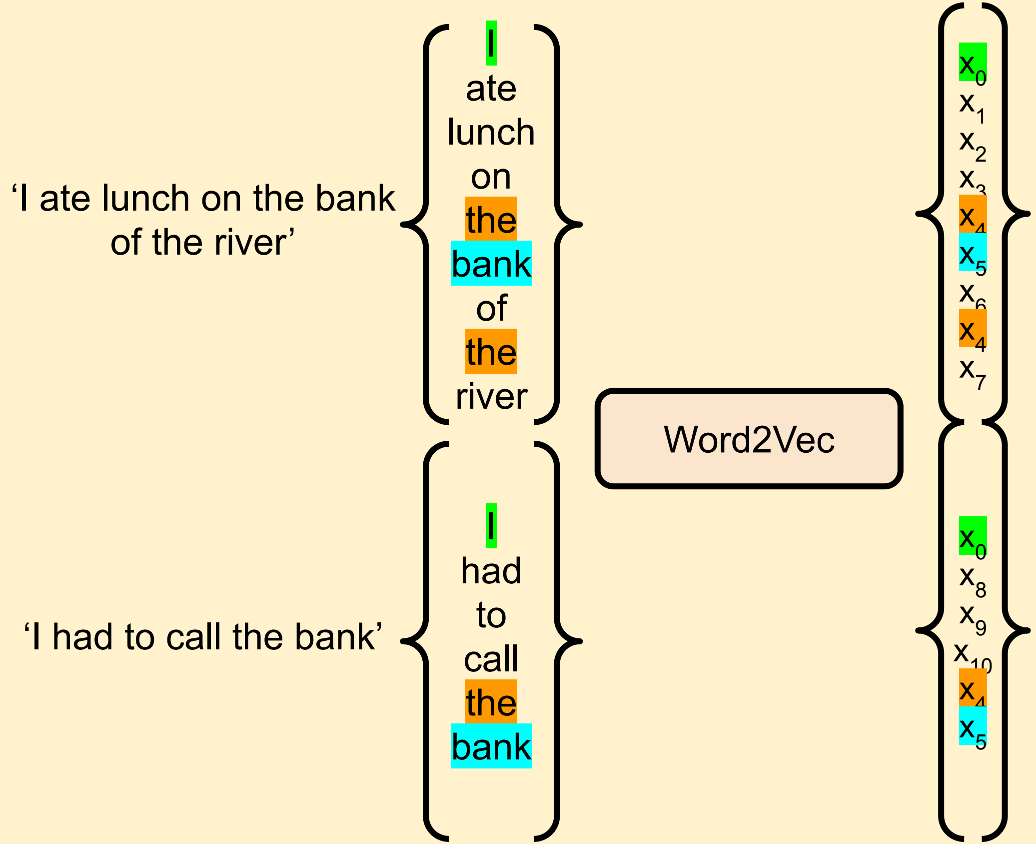
x_0
x_8
x_9
x_{10}
x_4
x_5

Static Embeddings: The Problem



Question
Does it matter if these share an embedding?

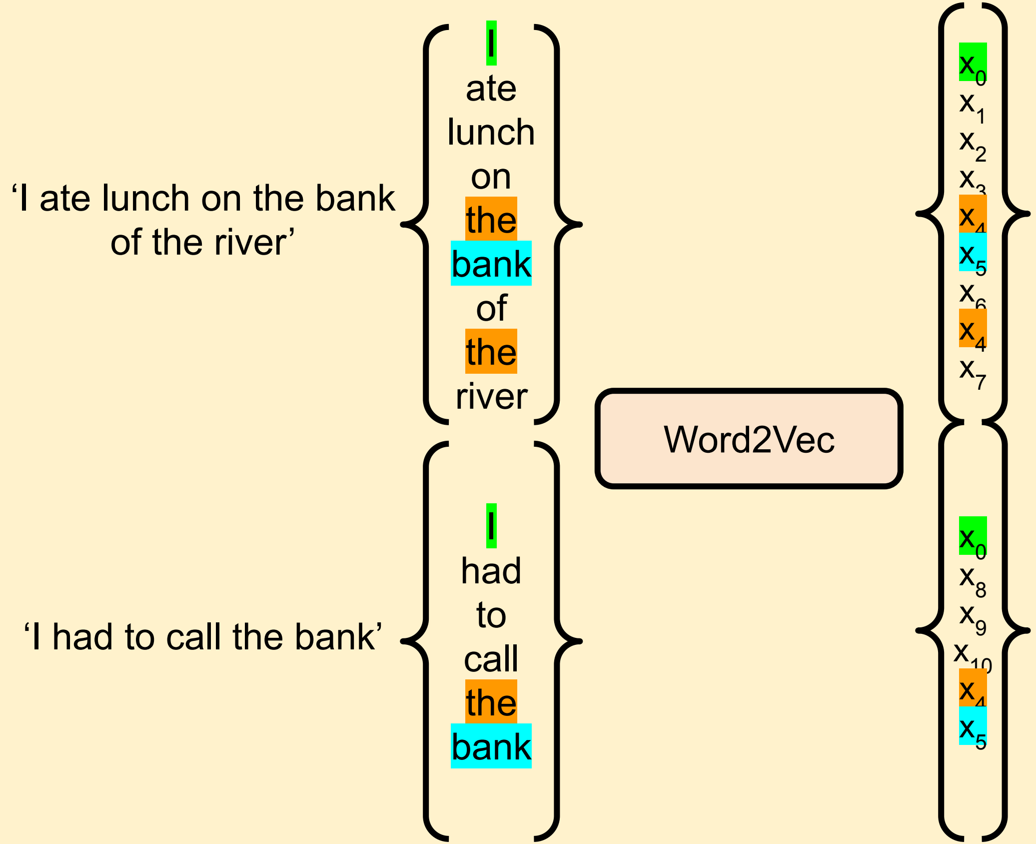
Static Embeddings: The Problem



Question
Are these embeddings effective in conveying meaning?

No, ‘bank’ has a different meaning in each document but shares an embedding.

Static Embeddings: The Problem



This is known as Polysemy!

Static Embeddings: The Problem

Context-dependent Sentiment

‘I *literally* can’t even right now.’
(Emphasis, not factual)
vs.
‘They *literally* ran a marathon.’
(Factual)

Static Embeddings: The Problem

Context-dependent Sentiment

'I *literally* can't even right now.'
(Emphasis, not factual)
vs.
'They *literally* ran a marathon.'
(Factual)

Grammatical Importance

'She has to *book* a meeting'
(Verb)
vs.
'He is reading a *book*.'
(Noun)

Static Embeddings: The Problem

Context-dependent Sentiment

‘I *literally* can’t even right now.’
(Emphasis, not factual)
vs.
‘They *literally* ran a marathon.’
(Factual)

Grammatical Importance

‘She has to *book* a meeting’
(Verb)
vs.
‘He is reading a *book*.’
(Noun)

Negation

‘This exam was not *good*.’
(‘not’ changes the ‘good’)
vs.
‘This exam was *good*’
(‘good’ is just ‘good’)

Static Embeddings: The Problem

Context-dependent Sentiment

‘I *literally* can’t even right now.’
(Emphasis, not factual)
vs.
‘They *literally* ran a marathon.’
(Factual)

Grammatical Importance

‘She has to *book* a meeting’
(Verb)
vs.
‘He is reading a *book*.’
(Noun)

Negation

‘This exam was not *good*.’
(‘not’ changes the ‘good’)
vs.
‘This exam was *good*’
(‘good’ is just ‘good’)

Domain-specific Meaning

‘Lucas trained an *model*.’
(machine learning model)
vs.
‘Lucas is a *model*.’
(Fashion model)

Static Embeddings: The Problem

Context-dependent Sentiment

'I *literally* can't even right now.'
(Emphasis, not factual)
vs.
'They *literally* ran a marathon.'
(Factual)

Grammatical Importance

'She has to *book* a meeting'
(Verb)
vs.
'He is reading a *book*.'
(Noun)

Negation

'This exam was not *good*.'
('not' changes the 'good')
vs.
'This exam was *good*'
('good' is just 'good')

Domain-specific Meaning

'Lucas trained an *model*.'
(machine learning model)
vs.
'Lucas is a *model*.'
(Fashion model)

Long-term Dependencies

'The apple wasn't put in the basket because it was too small.'

What was too small?

Static Embeddings: The Problem

Context-dependent Sentiment

‘I *literally* can’t even right now.’
(Emphasis, not factual)
vs.
‘They *literally* ran a marathon.’
(Factual)

Grammatical Importance

‘She has to *book* a meeting’
(Verb)
vs.
‘He is reading a *book*.’
(Noun)

Negation

These tie back to the ambiguities we discussed in Lecture 1.

See if you can match them up.

Domain-specific Meaning

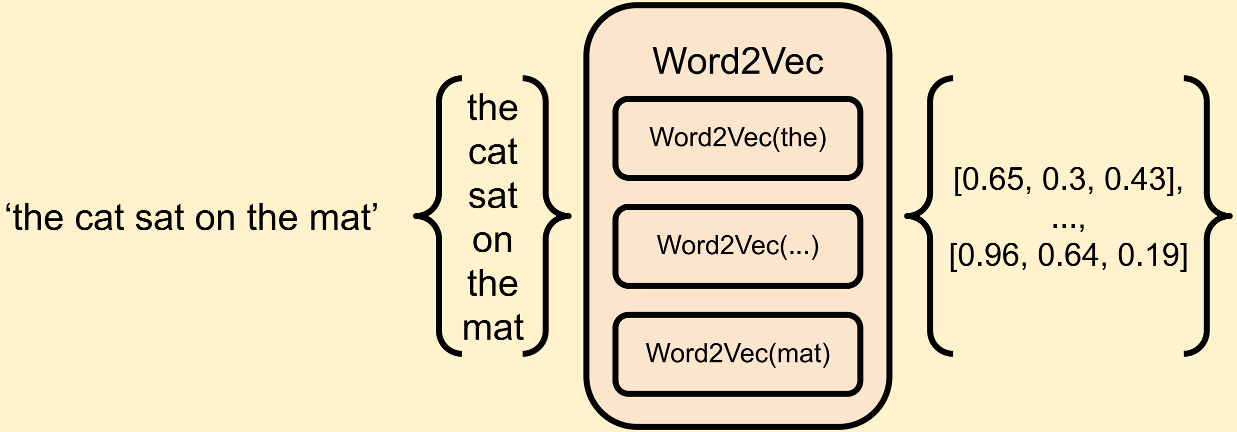
‘Lucas trained an *model*.’
(machine learning model)
vs.
‘Lucas is a *model*.’
(Fashion model)

Long-term Dependencies

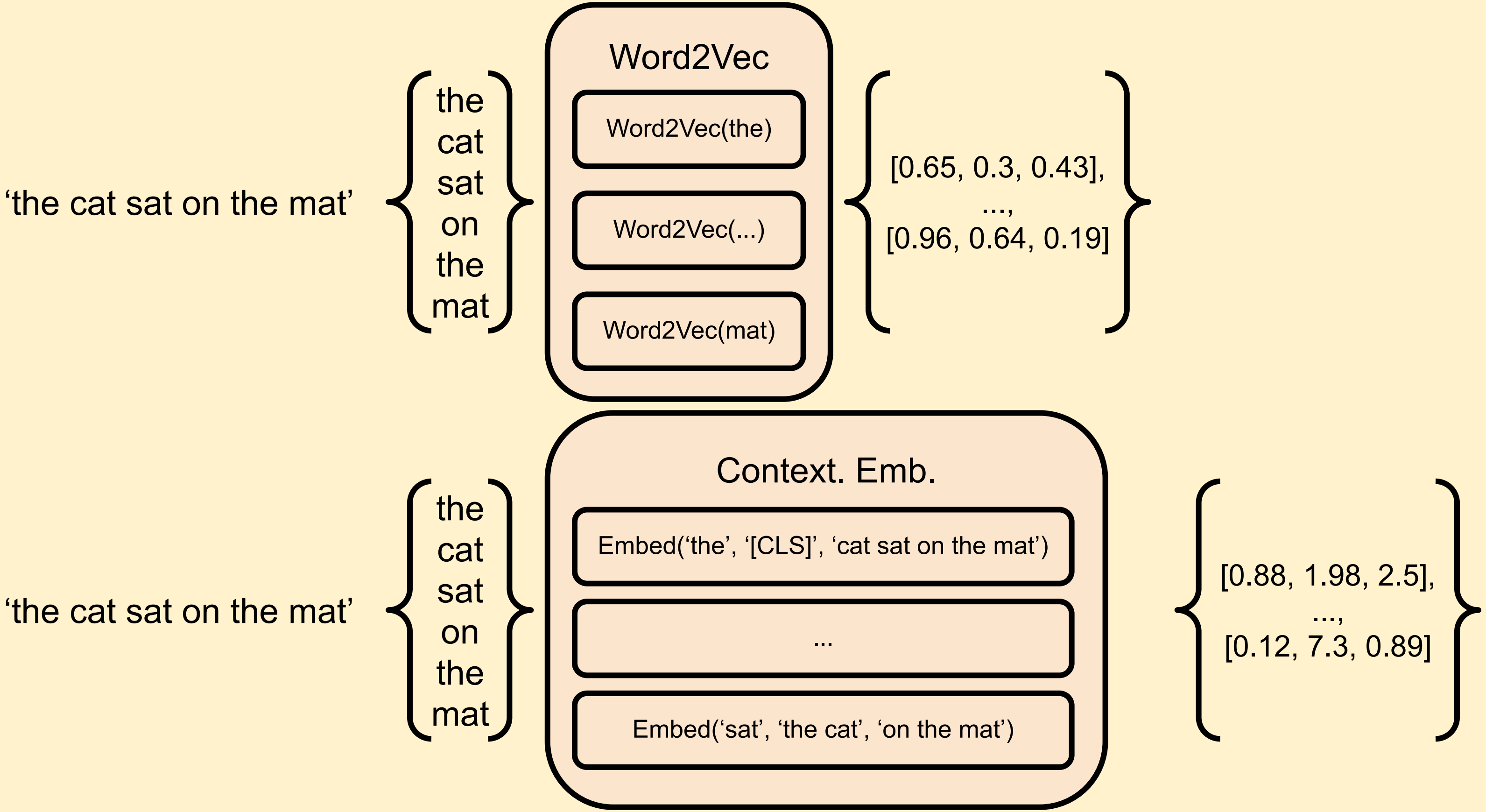
‘The apple wasn’t put in the basket because it was too small.’

What was too small?

Contextual Embeddings



Contextual Embeddings

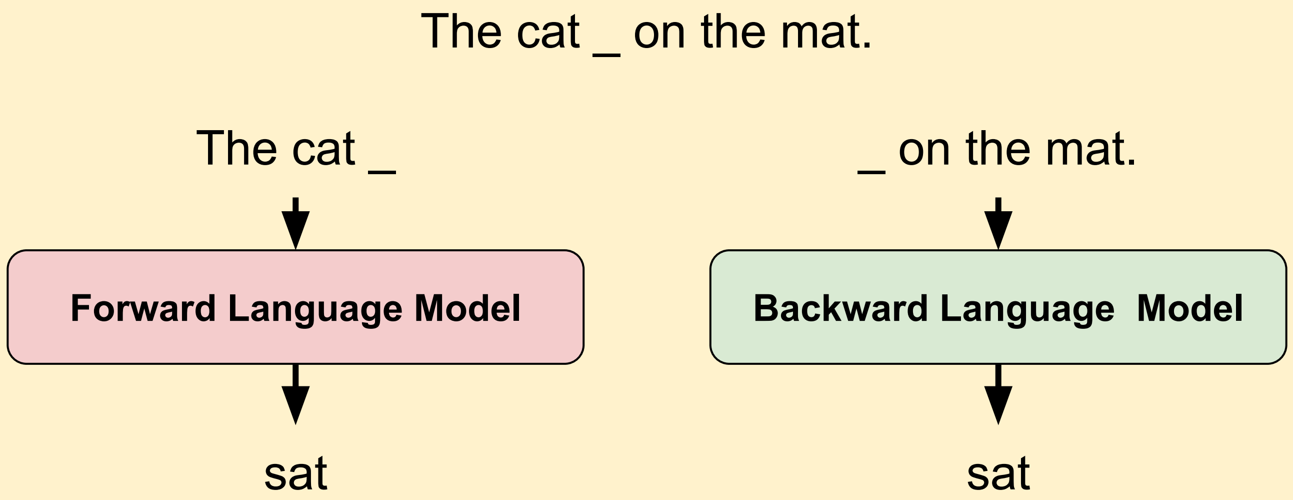


Contextual Embeddings: ELMO

- Using forward and backward word prediction to generate embeddings.

Contextual Embeddings: ELMo’s Architecture

- Using forward and backward word prediction to generate embeddings.

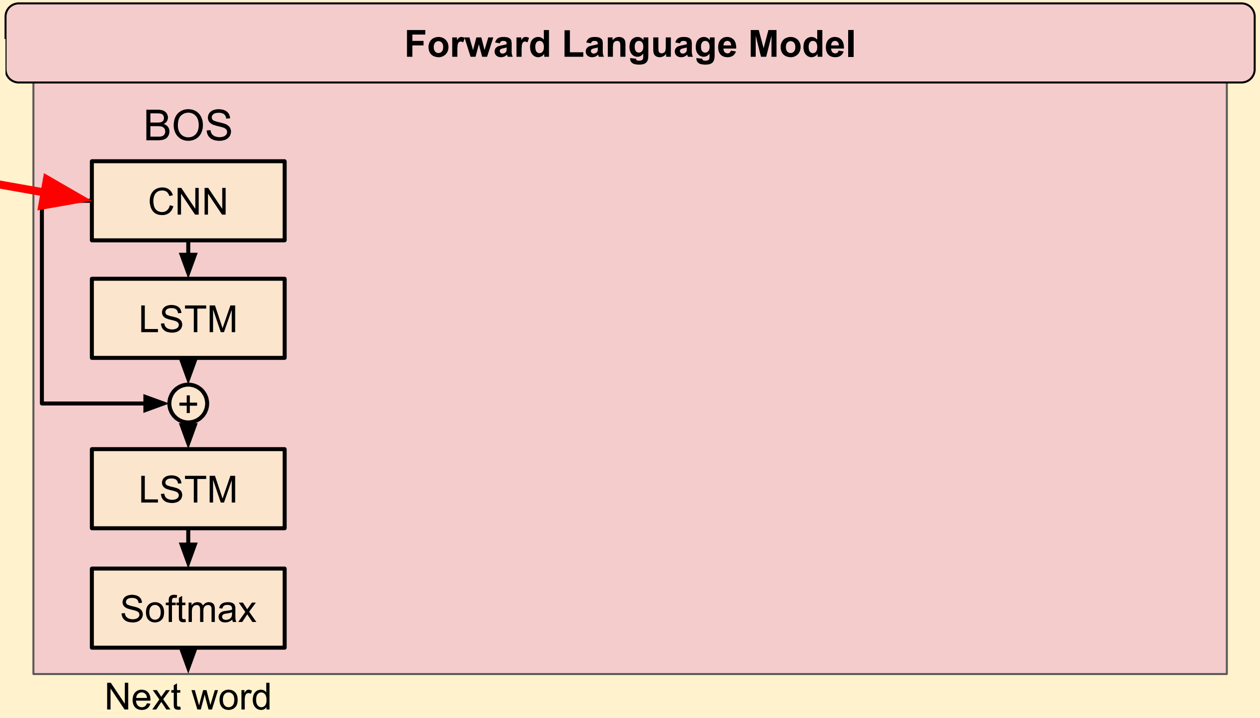


Contextual Embeddings: ELMo’s Architecture

- Using forward and backward word prediction to generate embeddings.

CNN-based character embedding.

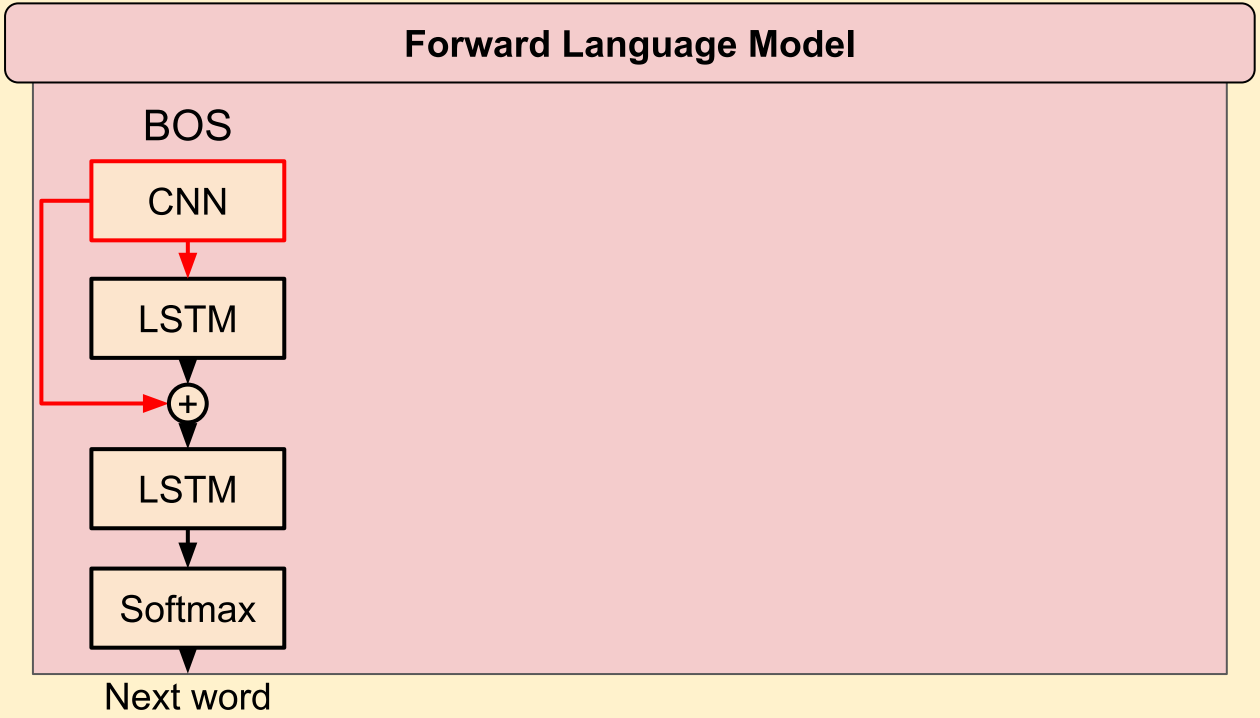
“The elephant ate the pasties”
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo's Architecture

- Using forward and backward word prediction to generate embeddings.

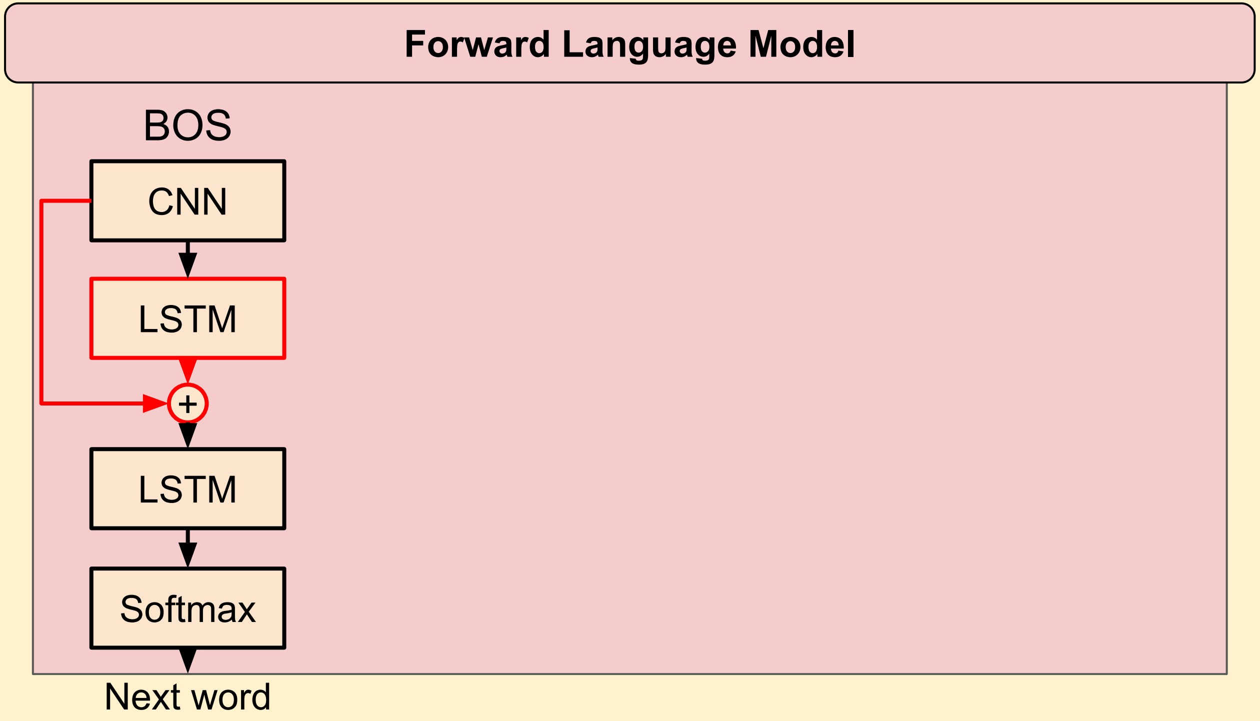
“The elephant ate the pasties”
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo’s Architecture

- Using forward and backward word prediction to generate embeddings.

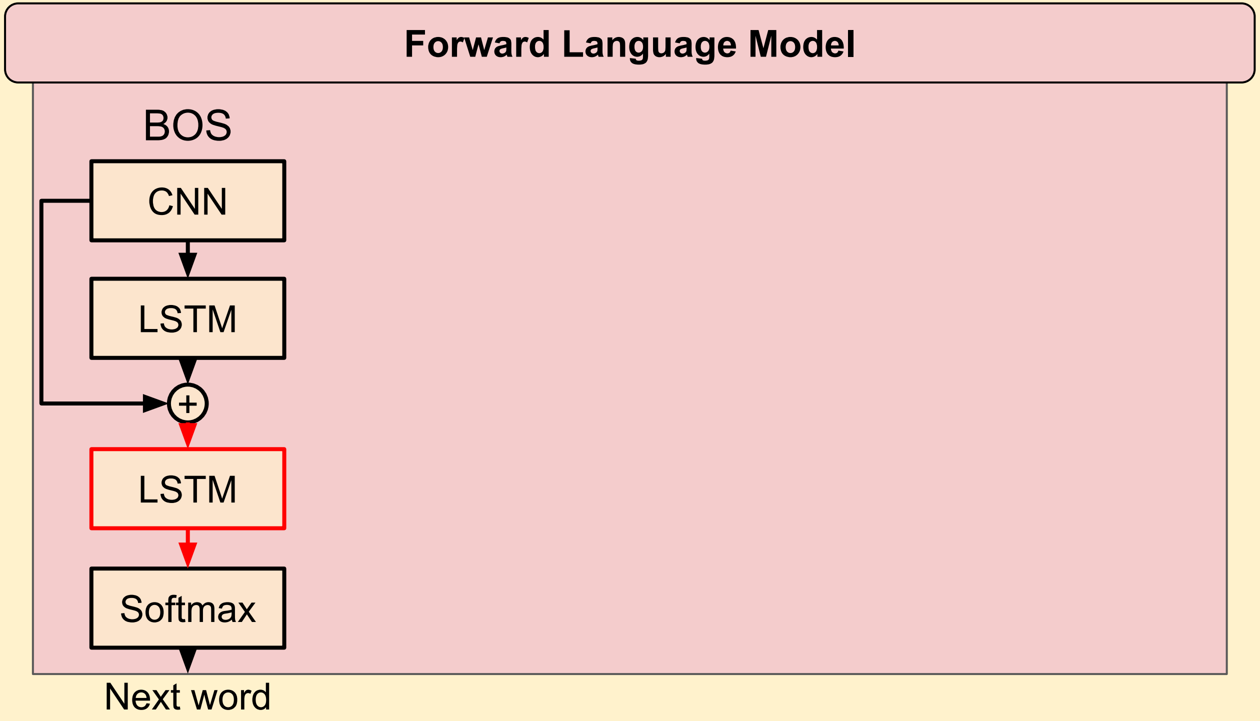
“The elephant ate the pasties”
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo’s Architecture

- Using forward and backward word prediction to generate embeddings.

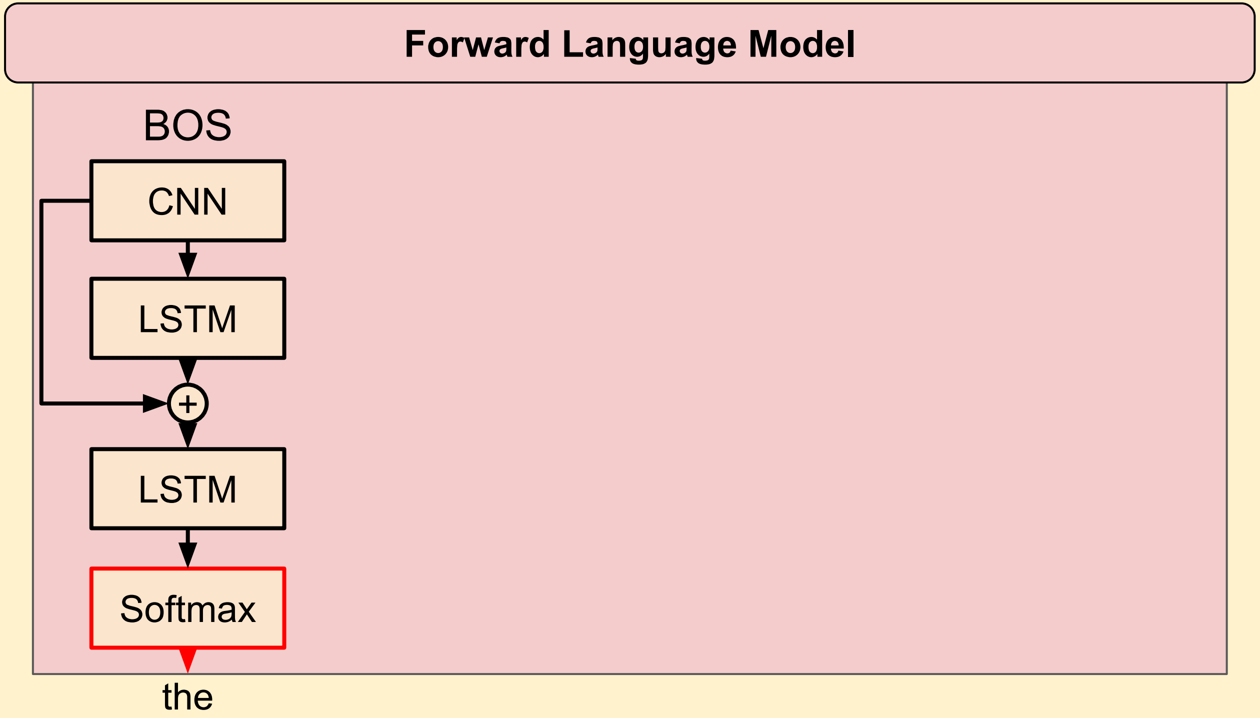
“The elephant ate the pasties”
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo's Architecture

- Using forward and backward word prediction to generate embeddings.

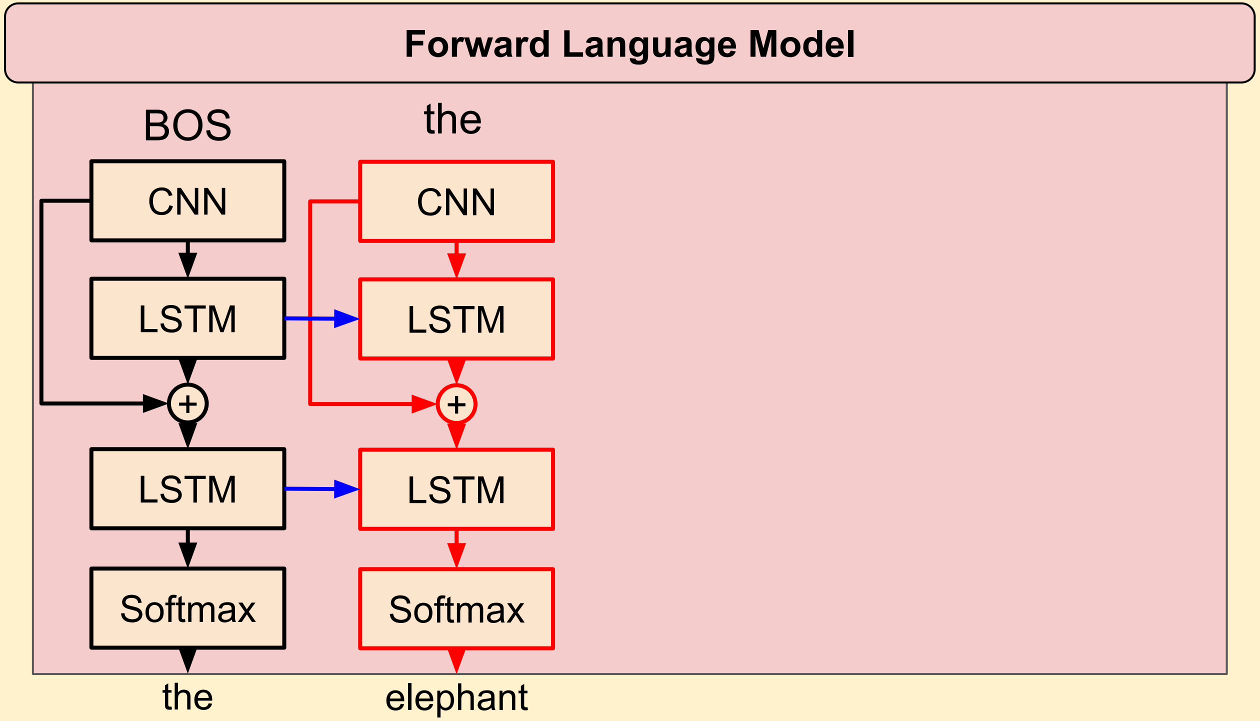
"The elephant ate the pasties"
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo's Architecture

- Using forward and backward word prediction to generate embeddings.

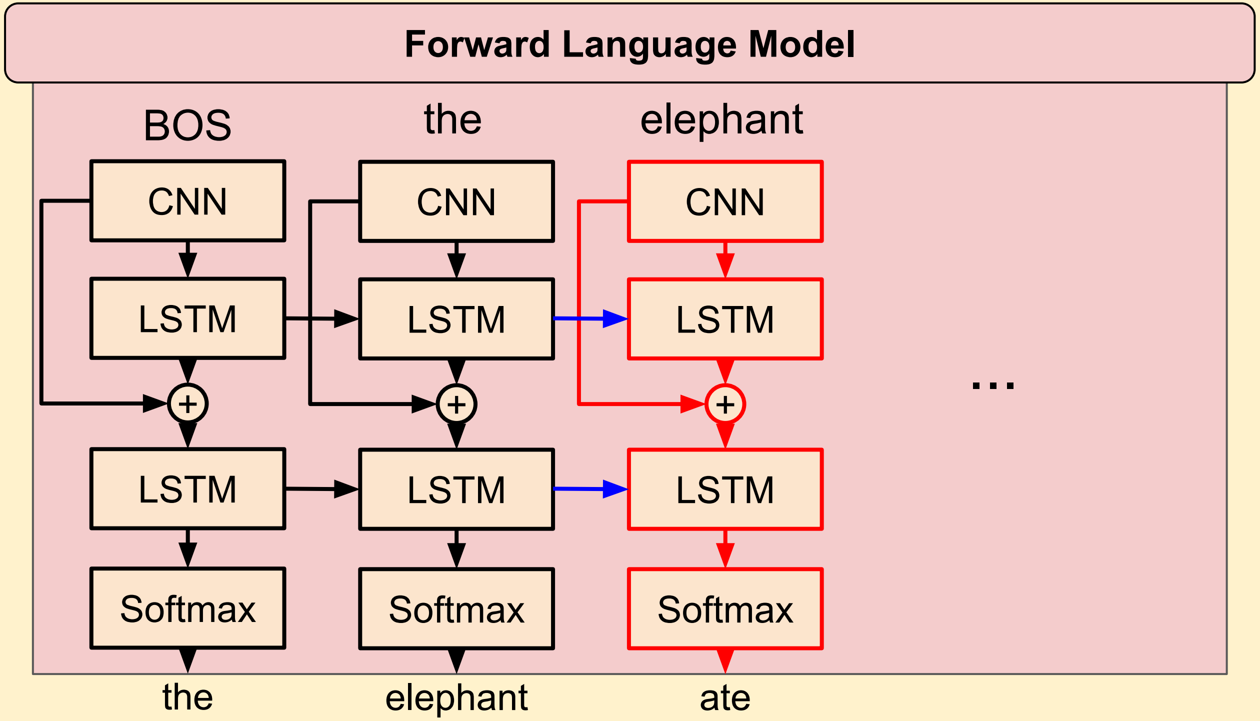
"The elephant ate the pasties"
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo's Architecture

- Using forward and backward word prediction to generate embeddings.

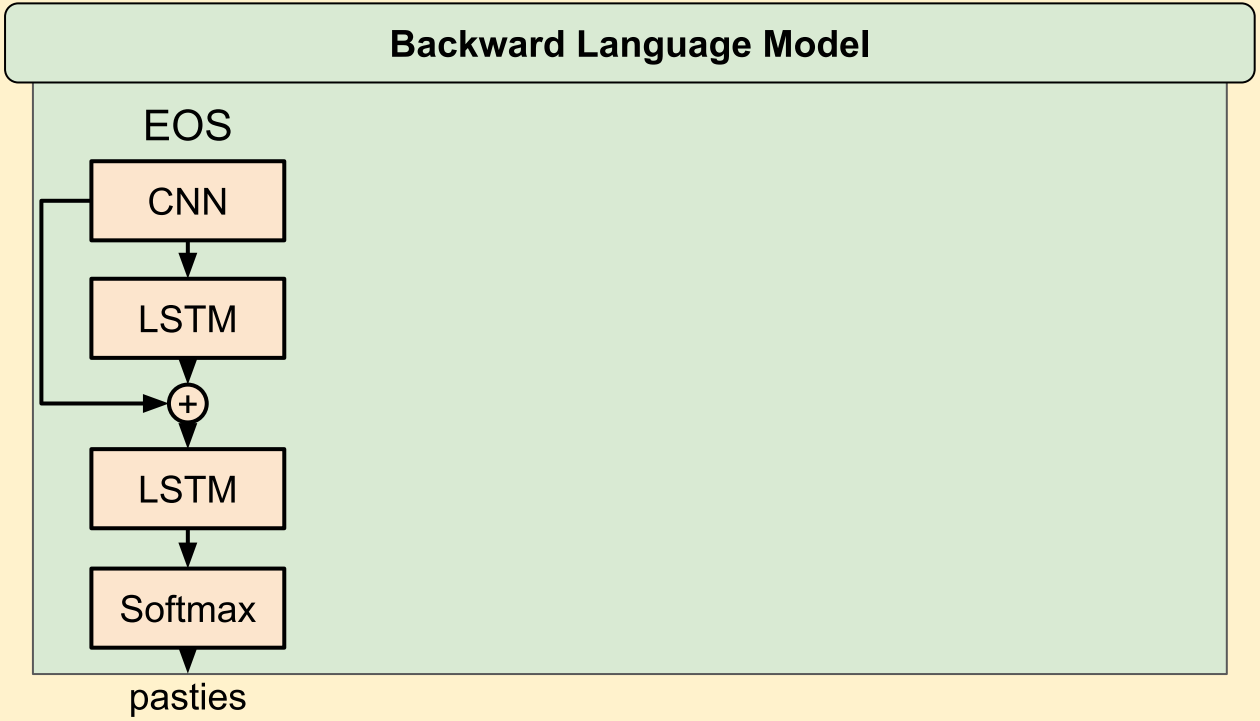
"The elephant ate the pasties"
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo's Architecture

- Using forward and backward word prediction to generate embeddings.

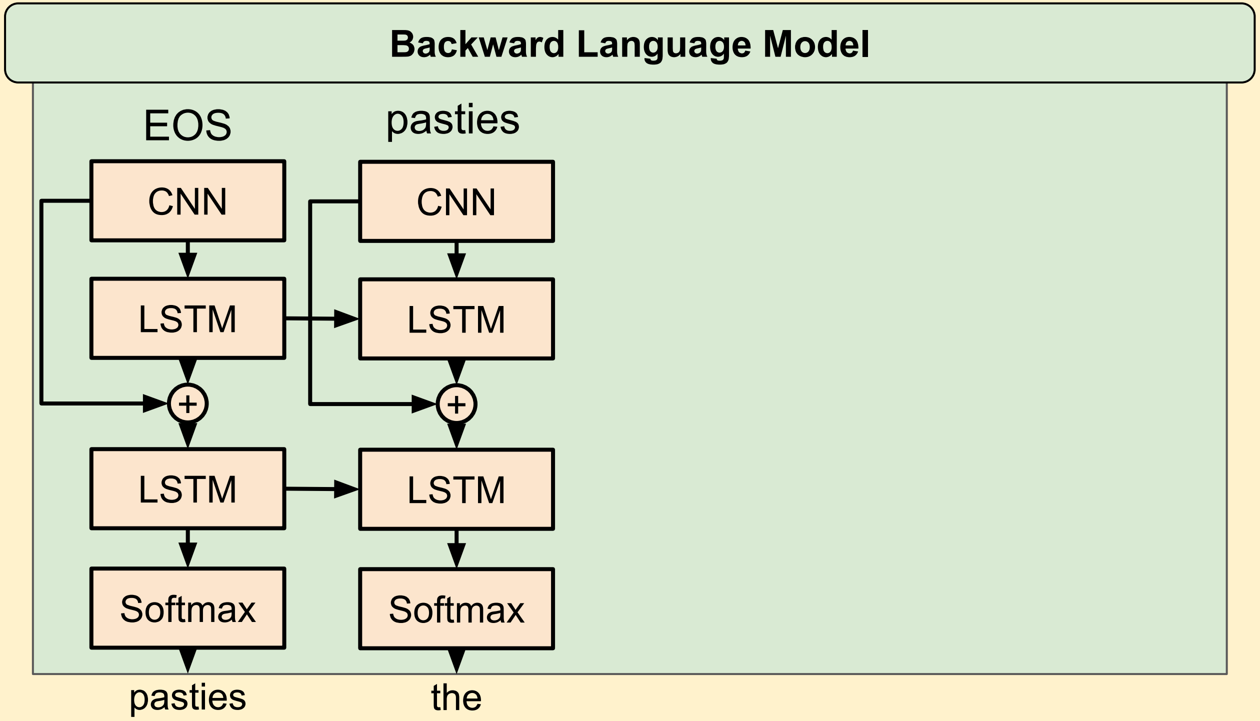
“The elephant ate the pasties”
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo’s Architecture

- Using forward and backward word prediction to generate embeddings.

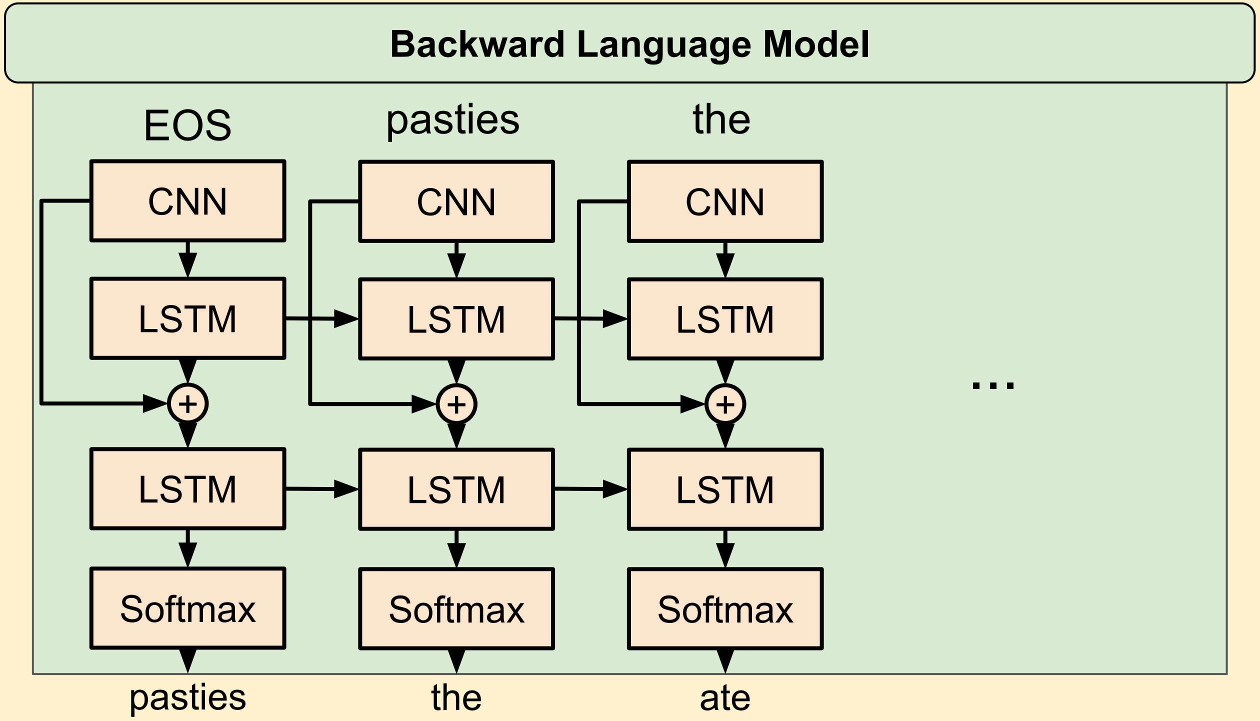
“The elephant ate the pasties”
[BOS, the, elephant, ate, the pasties, EOS]



Contextual Embeddings: ELMo's Architecture

- Using forward and backward word prediction to generate embeddings.

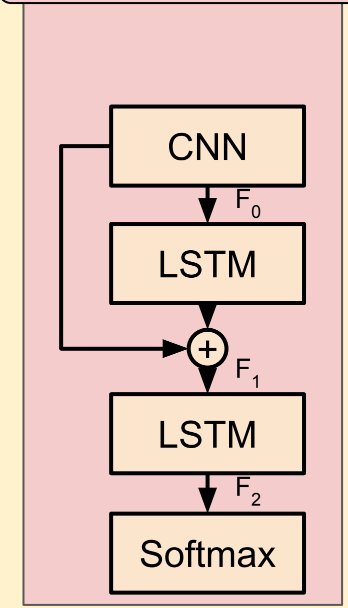
"The elephant ate the pasties"
[BOS, the, elephant, ate, the pasties, EOS]



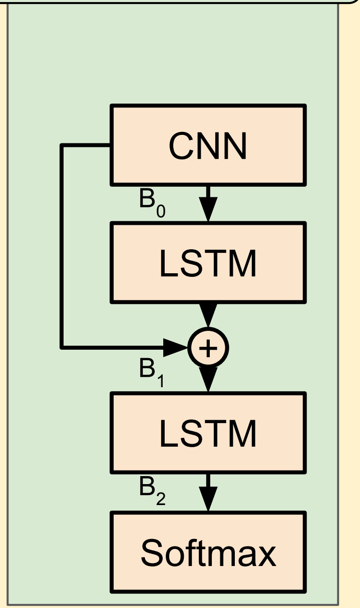
Contextual Embeddings: ELMo’s Architecture

- Using forward and backward word prediction to generate embeddings.

Forward Language Model



Backward Language Model

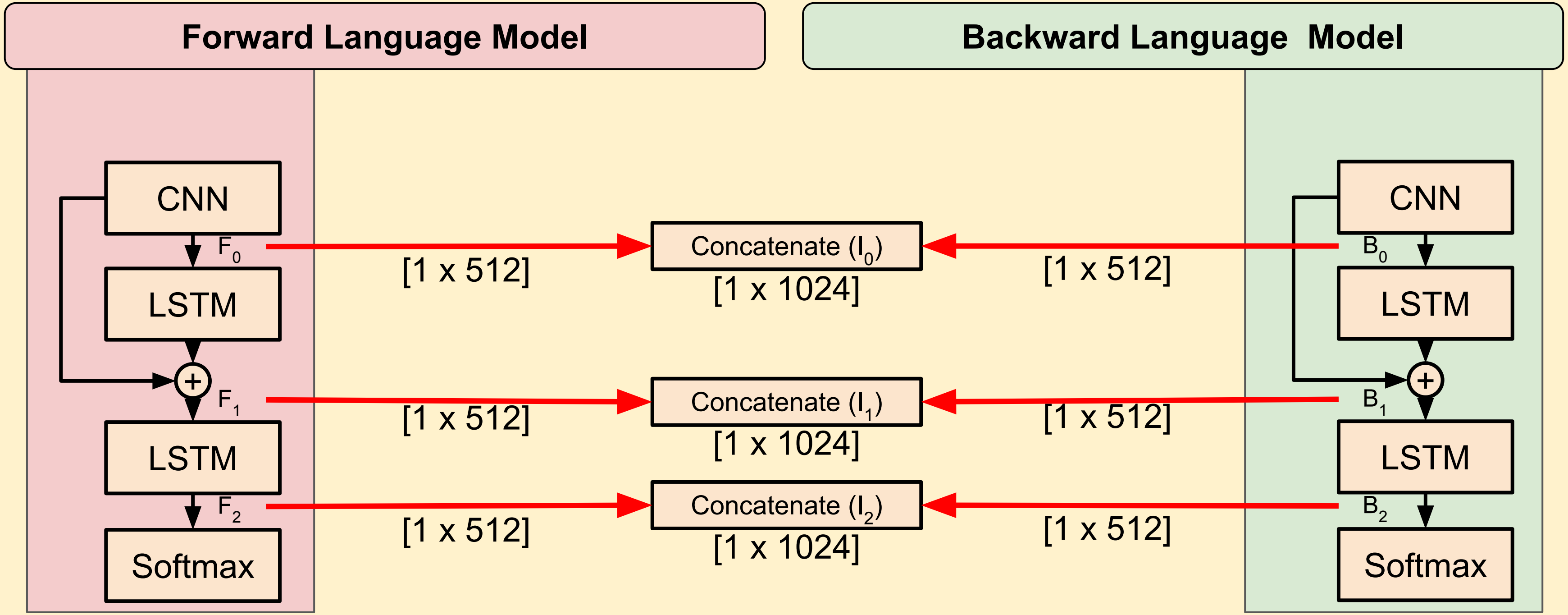


Question
What’s the benefit of using both forward and backward LM as opposed to just a forward LM?

BONUS: Can you give an example of a document in which a forward LM wouldn’t be sufficient.

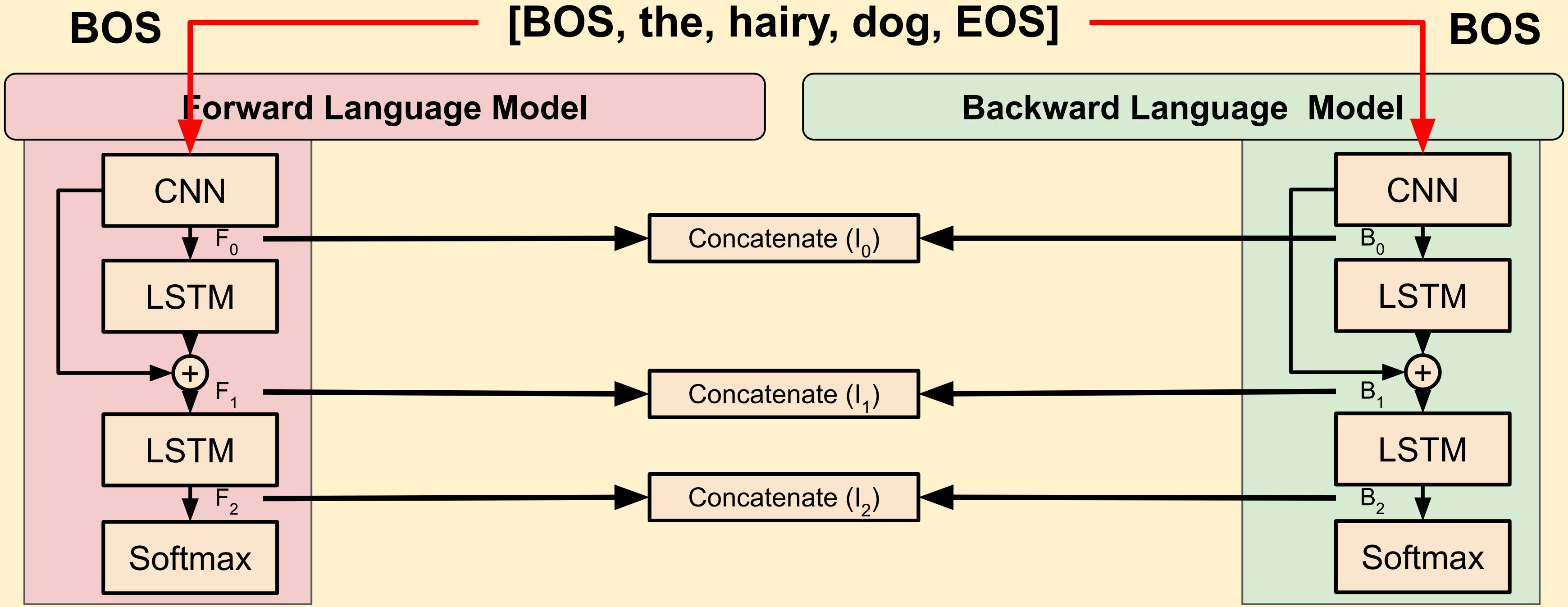
Contextual Embeddings: ELMo's Embeddings

- Using forward and backward word prediction to generate embeddings.



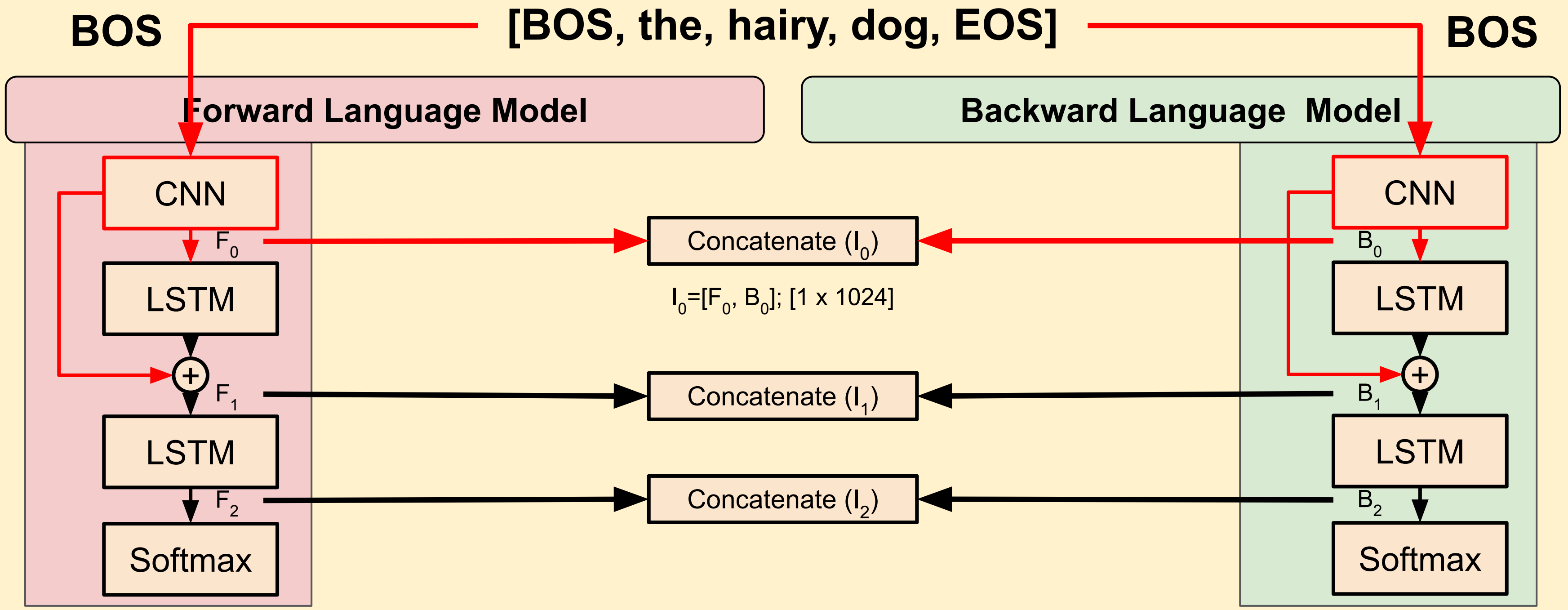
Contextual Embeddings: ELMo's Embeddings

- Using forward and backward word prediction to generate embeddings.
- Internal states can be used to generate embeddings.



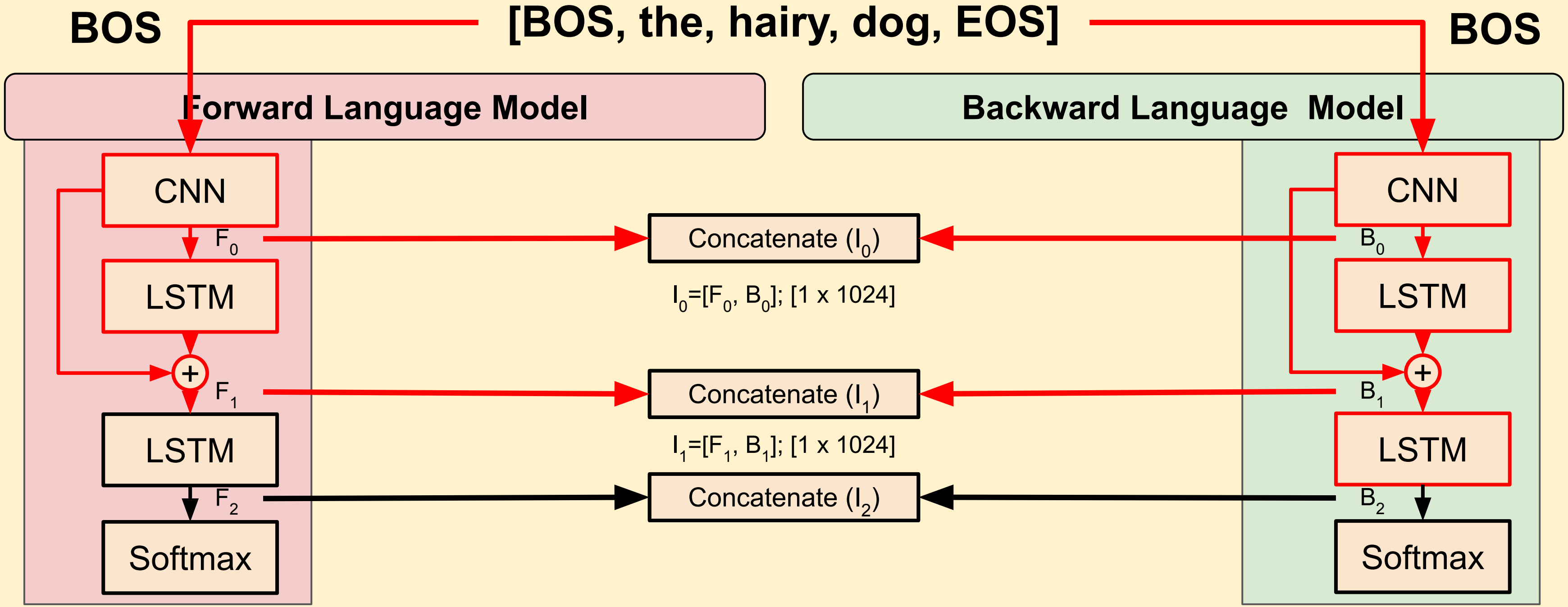
Contextual Embeddings: ELMo's Embeddings

- Using forward and backward word prediction to generate embeddings.
- Internal states can be used to generate embeddings.



Contextual Embeddings: ELMo's Embeddings

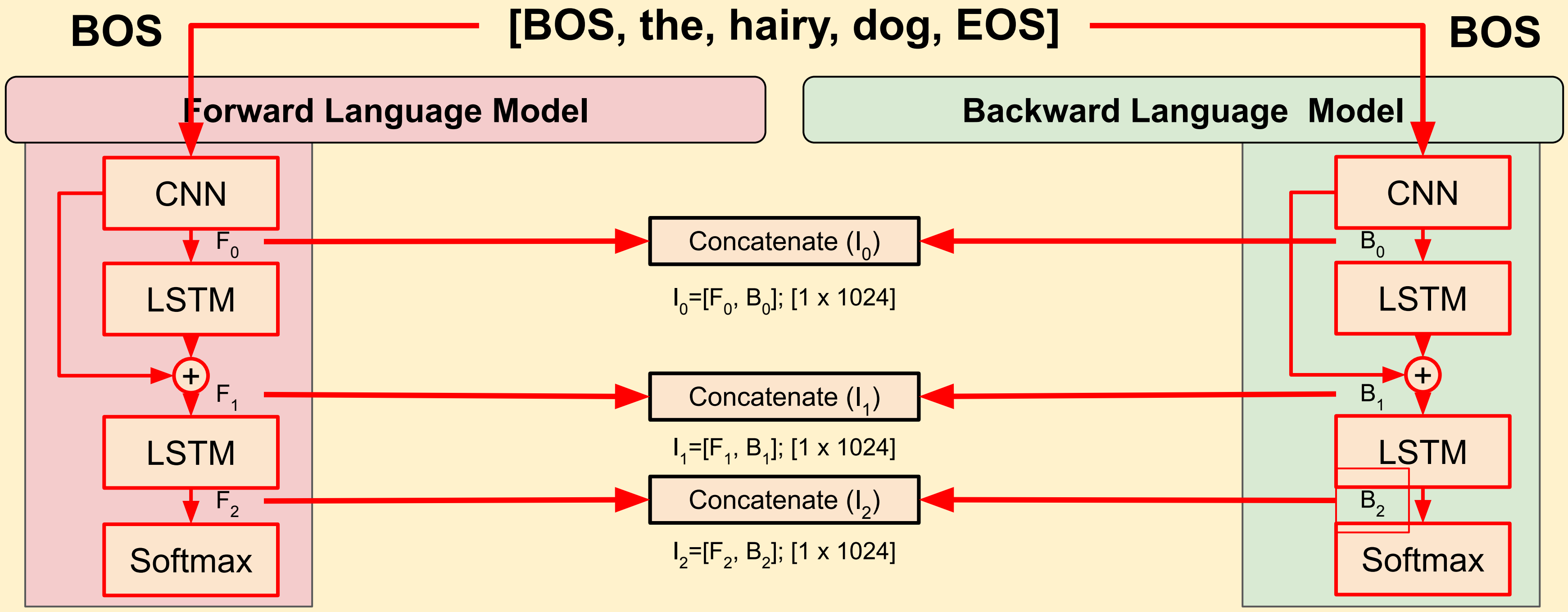
- Using forward and backward word prediction to generate embeddings.
- Internal states can be used to generate embeddings.



Contextual Embeddings: ELMo's Em

Question
Why would multiple internal states be useful?

- Using forward and backward word prediction to g
- Internal states can be used to generate embeddings.



Contextual Embeddings: ELMo's Embeddings

- Using forward and backward word prediction to generate embeddings.
- Internal states can be used to generate embeddings.

[BOS, the, hairy, dog, EOS]

BOS →

Concatenate (l_0)	l_0=[F_0, B_0]; [1 x 1024]
Concatenate (l_1)	l_1=[F_1, B_1]; [1 x 1024]
Concatenate (l_2)	l_2=[F_2, B_2]; [1 x 1024]

'PreTrained Embeddings'

- This BiLSTM should be trained on large, generic corpora.
- ELMo offers a way to make the embeddings task specific.

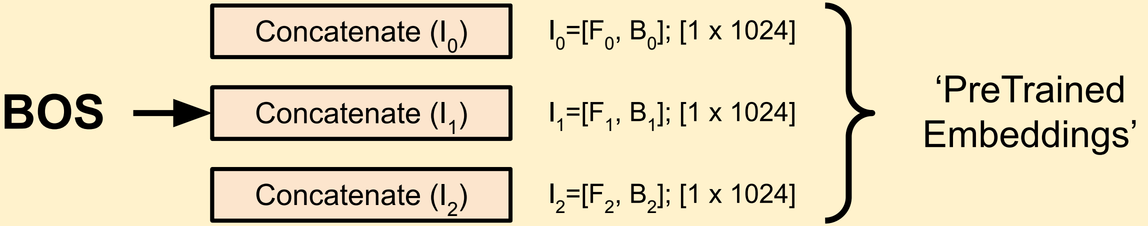
Contextual Embeddings:

Question

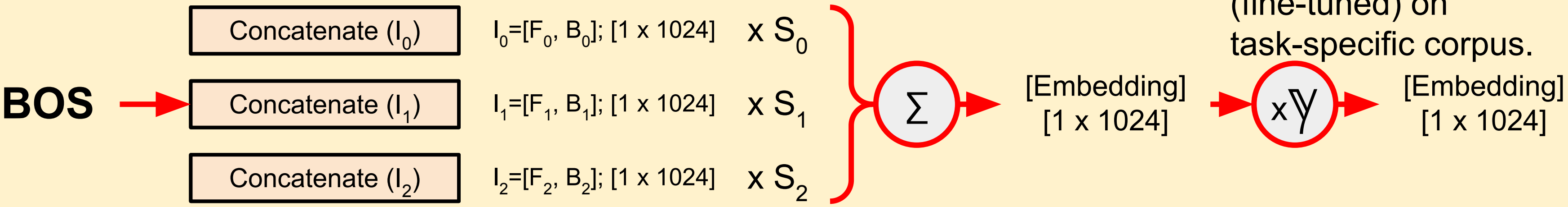
Thinking back to the previous question, why might you want to weight the internal states for a specific task?

- Using forward and backward word pred...
- Internal states can be used to generate embeddings.

[BOS, the, hairy, dog, EOS]



- This BiLSTM should be trained on large, generic corpora.
- ELMo offers a way to make the embeddings task specific.
- S_n and γ trained (fine-tuned) on task-specific corpus.



Contextual Embeddings: ELMo's Architecture

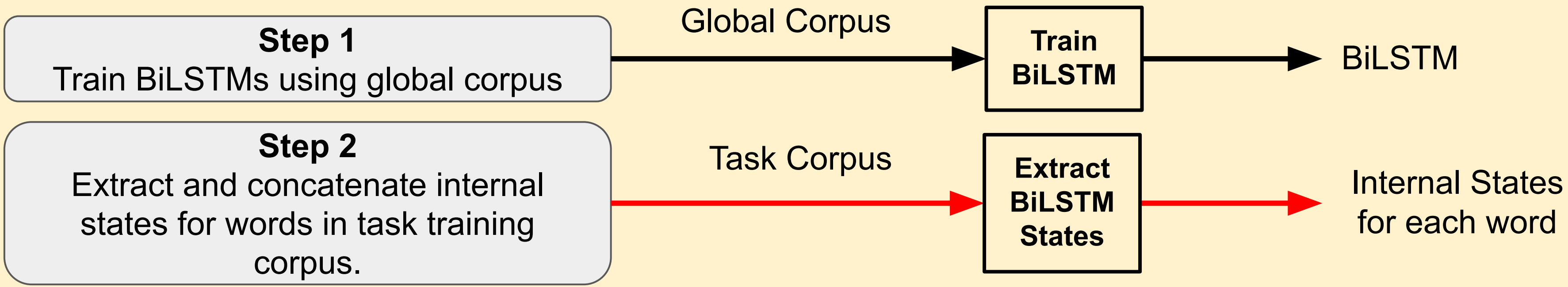
- Let's take a step back and look at this process holistically with a use-case:
  - We want to generate a sentiment classification model.
  - We have a corpus of documents provided with labels (task corpus).
  - We have a corpus of global documents (e.g. 1.9TB Reddit Corpora) (global corpus).

Contextual Embeddings: ELMo's Architecture

- Let's take a step back and look at this process holistically with a use-case:
  - We want to generate a sentiment classification model.
  - We have a corpus of documents provided with labels (task corpus).
  - We have a corpus of global documents (e.g. 1.9TB Reddit Corpora) (global corpus).

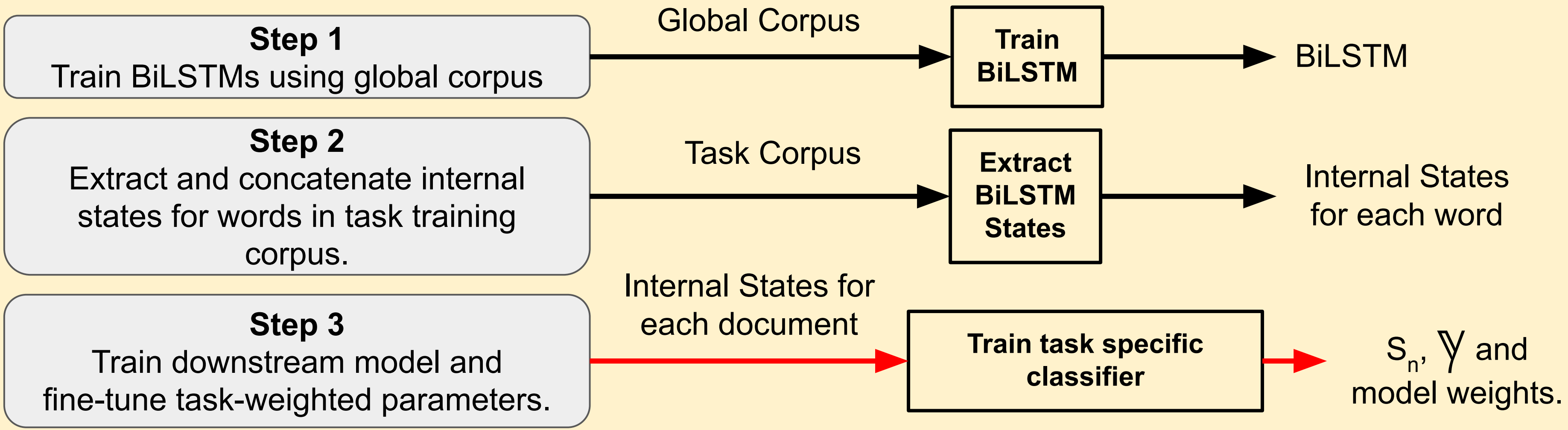
Contextual Embeddings: ELMo's Architecture

- Let's take a step back and look at this process holistically with a use-case:
  - We want to generate a sentiment classification model.
  - We have a corpus of documents provided with labels (task corpus).
  - We have a corpus of global documents (e.g. 1.9TB Reddit Corpora) (global corpus).

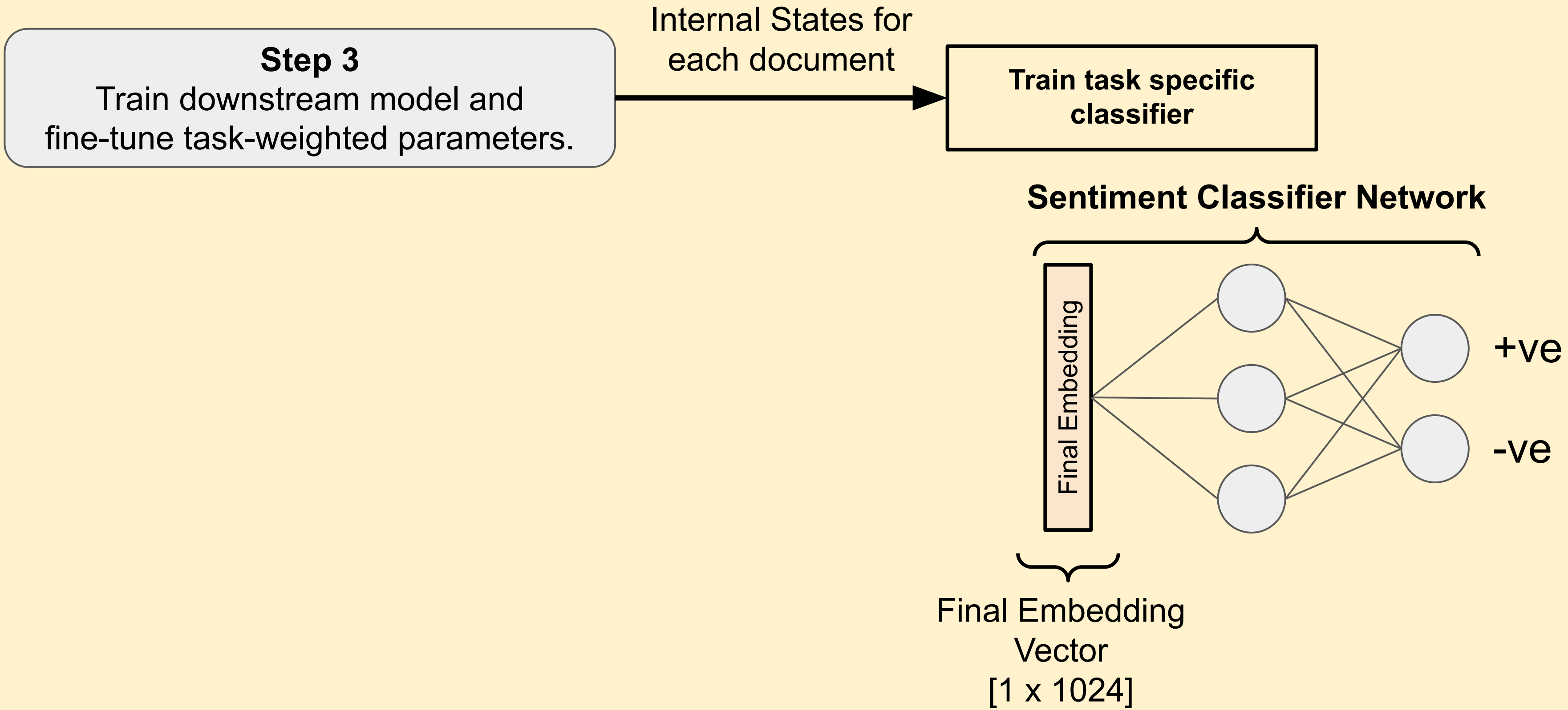


Contextual Embeddings: ELMo’s Architecture

- Let’s take a step back and look at this process holistically with a use-case:
  - We want to generate a sentiment classification model.
  - We have a corpus of documents provided with labels (task corpus).
  - We have a corpus of global documents (e.g. 1.9TB Reddit Corpora) (global corpus).



Contextual Embeddings: ELMo's Architecture



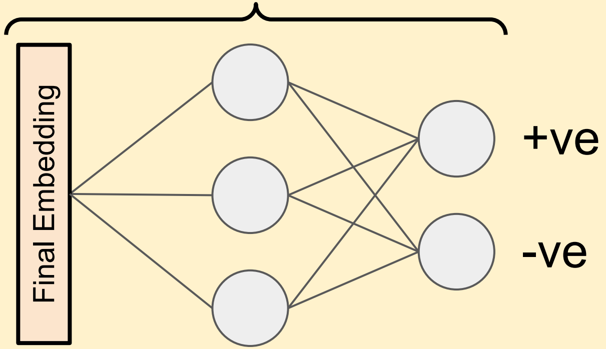
Contextual Embeddings: ELMo's Architecture

Step 3
Train downstream model and fine-tune task-weighted parameters.

Internal States for each document

Train task specific classifier

Sentiment Classifier Network



Final Embedding Vector
[1 x 1024]

Internal State 0

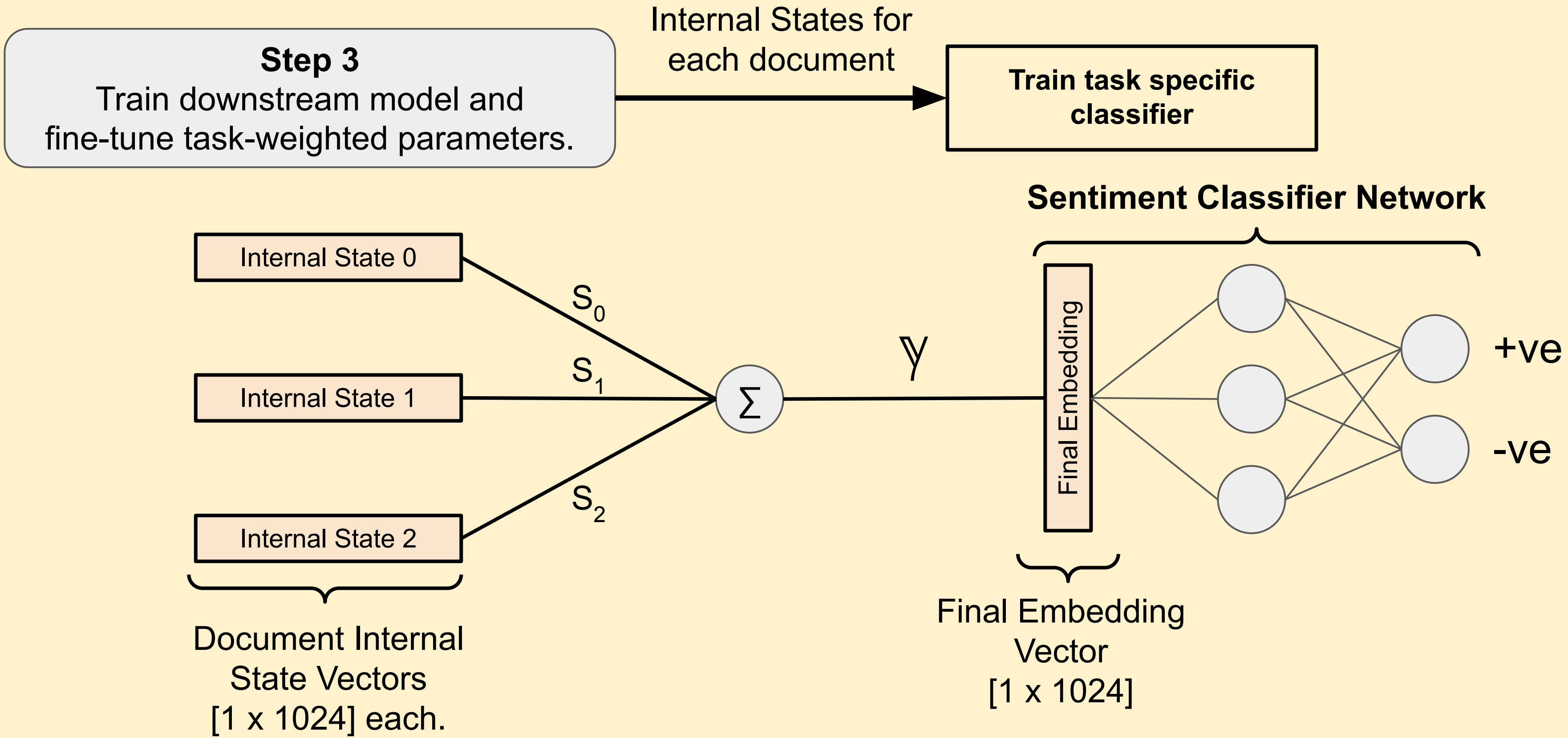
Internal State 1

Internal State 2

Document Internal State Vectors
[1 x 1024] each.

[1 x 512] for forward
[1 x 512] for backward

Contextual Embeddings: ELMo’s Architecture

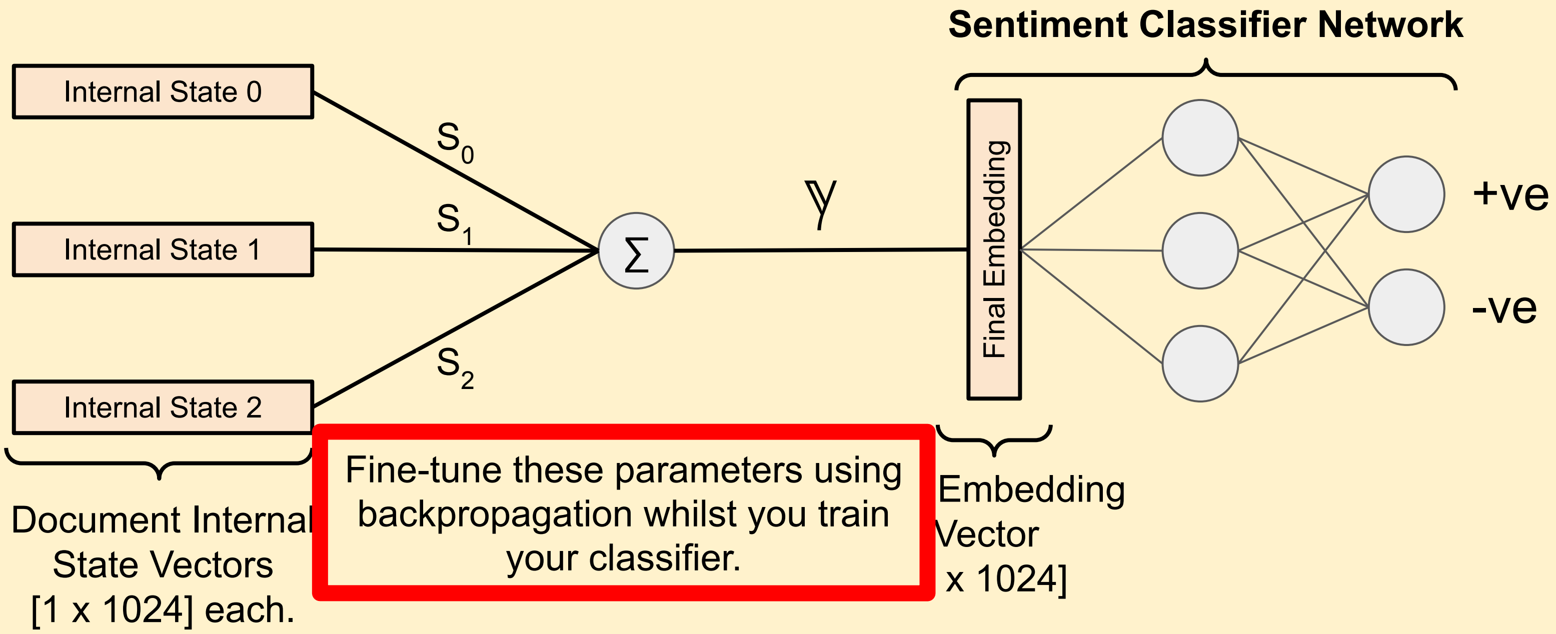


Contextual Embeddings: ELMo's Architecture

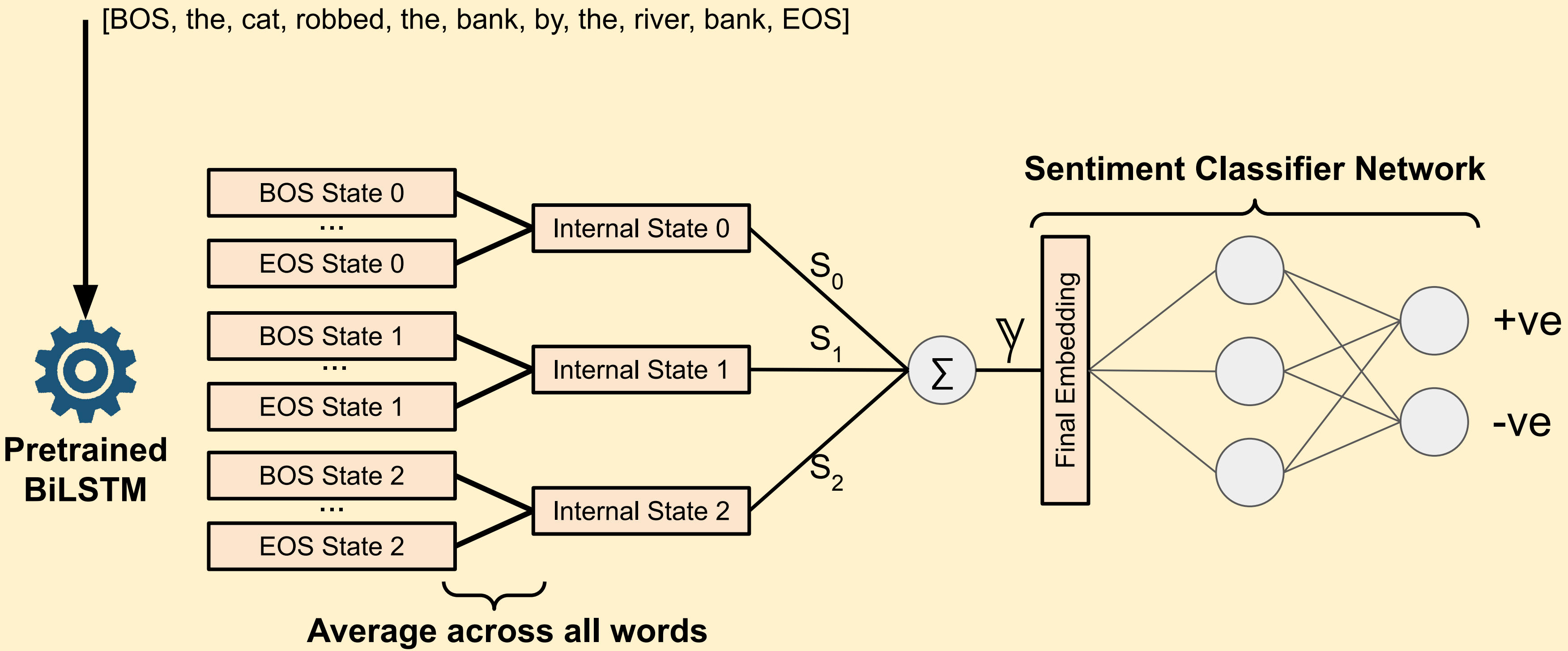
Step 3
Train downstream model and
fine-tune task-weighted parameters.

Internal States for
each document

**Train task specific
classifier**



Contextual Embeddings: ELMo’s Inference



Contextual Embeddings: ELMo’s Inference

- Do we need to train a BiLSTM everytime we want to use ELMo?
    - No! And that’s the beauty of it.

Contextual Embeddings: ELMo’s Inference

- Do we need to train a BiLSTM everytime we want to use ELMo?
    - No! And that’s the beauty of it.

- Researchers and organizations have already pre trained BiLSTMs for you to use using datasets far larger than the Reddit corpus.
    - Google, Microsoft, OpenAI etc.
    - So you only need to fine-tune the context parameters and your model.

Contextual Embeddings: ELMo's Inference

- Do we need to train a BiLSTM everytime we want to use ELMo?
  - No! And that's the beauty of it.

- Researchers and organizations have already pre trained BiLSTMs for you to use using datasets far larger than the Reddit corpus.
  - Google, Microsoft, OpenAI etc.
  - So you only need to fine-tune the context parameters and your model.

Use an off-the-shelf solution where possible.

Pretrained BiLSTM

ELMo Context Parameters

Task-specific Model

PreTrained Embeddings

- Why do we use pre-trained embeddings?

Linguistic Generalisation

Training the initial embeddings on a large, general corpus allows the model to capture global language components and mitigates overfitting to a given task.

PreTrained Embeddings

- Why do we use pre-trained embeddings?

Linguistic Generalisation

Training the initial embeddings on a large, general corpus allows the model to capture global language components and mitigates overfitting to a given task.

Question

Can you name three language components we saw in Lecture 1?

PreTrained Embeddings

- Why do we use pre-trained embeddings?

Linguistic Generalisation

Training the initial embeddings on a large, general corpus allows the model to capture global language components and mitigates overfitting to a given task.

Reduced Data Needs

Using a ‘frozen’ weights not only reduces the data required to train the model but also increases speed of convergence in the downstream task.

PreTrained Embeddings

- Why do we use pre-trained embeddings?

Linguistic Generalisation

Training the initial embeddings on a large, general corpus allows the model to capture global language components and mitigates overfitting to a given task.

Reduced Data Needs

Using a ‘frozen’ weights not only reduces the data required to train the model but also increases speed of convergence in the downstream task.

Modularisation

Pretrained embeddings allow a ‘plug and play’ approach to NLP tasks without the overhead of extensive deep learning training regimes.

PreTrained Embeddings

- What are the disadvantages?

Domain-specific Meaning

‘Lucas trained an *model*.’
vs.
‘Lucas is a *model*.’

(See start of lecture)

Bias and Fairness

No control over the original training corpus, therefore any embedded bias in the pretrained model will be present in the fine-tuned task model.

Computational Efficiency

Even without training a deep embedding model the pre-trained alternatives come with high memory costs and can have long inference times.

https://blog.google/products-and-platforms/products/search/search-language-understanding-bert/

Use-Case: Google Search (IR)

- Google wanted to enhance their search functionality by improving the natural language understanding of queries and pages.

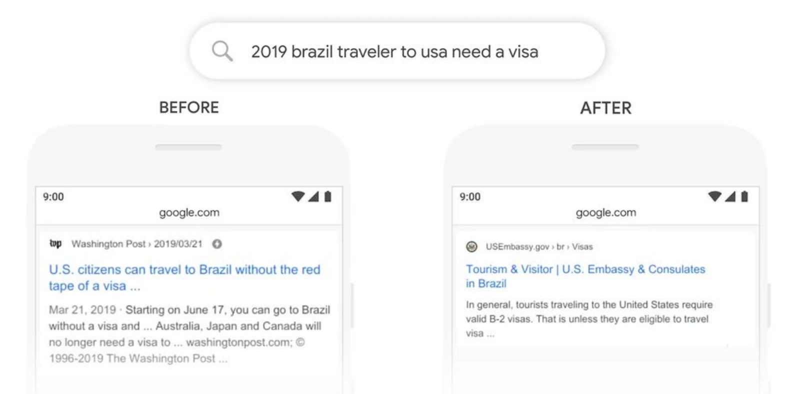
- They used a contextual embedding called BERT which we will visit in week 5 (some of you may remember this from Deep Learning with Vinay).
    - BERT embedded search queries and documents to improve similarity matching.

https://blog.google/products-and-platforms/products/search/search-language-understanding-bert/

Use-Case: Google Search (IR)

- Google wanted to enhance their search functionality by improving the natural language understanding of queries and pages.

- They used a contextual embedding called BERT which we will visit in week 5 (some of you may remember this from Deep Learning with Vinay).
    - BERT embedded search queries and documents to improve similarity matching.

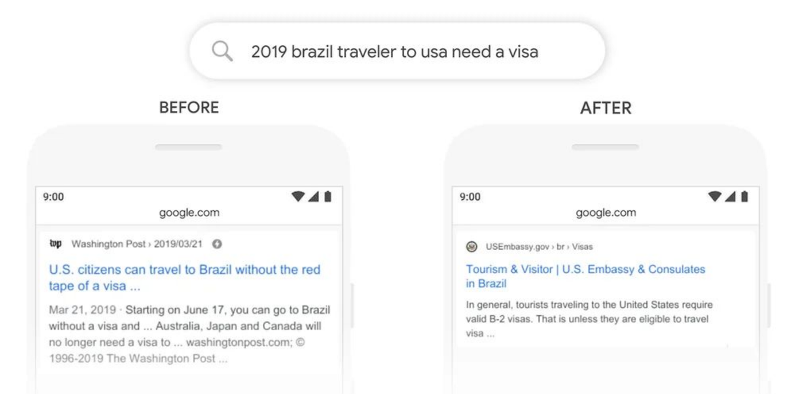


https://blog.google/products-and-platforms/products/search/search-language-understanding-bert/

Use-Case: Google Search (IR)

- Google wanted to enhance their search functionality by improving the natural language understanding of queries and pages.

- They used a contextual embedding called BERT which we will visit in week 5 (some of you may remember this from Deep Learning with Vinay).
    - BERT embedded search queries and documents to improve similarity matching.



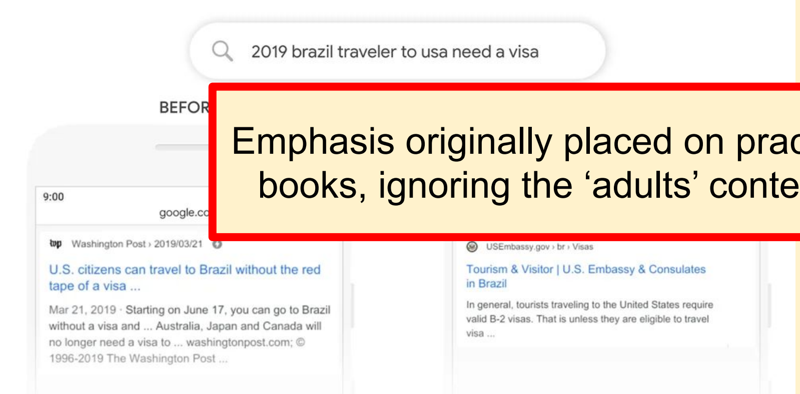
Brazilian tourist visiting the US, not the other way around. It's the contextual 'to' which indicates this.

https://blog.google/products-and-platforms/products/search/search-language-understanding-bert/

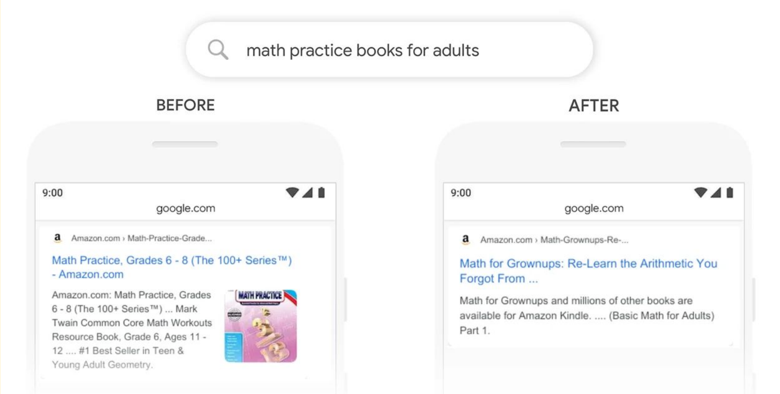
Use-Case: Google Search (IR)

- Google wanted to enhance their search functionality by improving the natural language understanding of queries and pages.

- They used a contextual embedding called BERT which we will visit in week 5 (some of you may remember this from Deep Learning with Vinay).
  - BERT embedded search queries and documents to improve similarity matching.



Emphasis originally placed on practice books, ignoring the 'adults' context.



Learning Objectives

Describe the limitations of static embeddings.

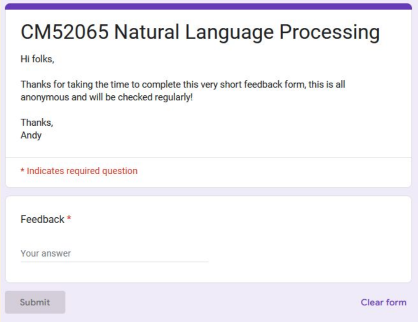
Illustrate the architecture of ELMo.

Explain the benefits and drawbacks of pre-trained weights.

Feedback

- We have 10 weeks together, let me know how it is going!
- This is my first time running this course, so I am keen to hear your thoughts.
- It's never too early or too late to provide feedback.

https://forms.gle/9WsY57cZKSkLQz8k6


CM52065 Natural Language Processing

Hi folks,

Thanks for taking the time to complete this very short feedback form, this is all anonymous and will be checked regularly!

Thanks,
Andy

* Indicates required question

Feedback *

Your answer

Submit

Clear form